Weekly update 6

Final campaign results: every checkpoint closed, all four research questions answered

Nandan Hemanth · 22 October 2026

Previously: what was in place before the final run

DRIFT IN ONE LINE Hold the agent's memory system fixed, change only the forgetting policy, and measure accuracy, staleness and cost.

- **Checkpoints set (Week 4):** best grace period, best retrieval weights, small vs frontier models, thesis revision, RQ4.
- **Weights searched (Week 5):** all 1,000 weightings on two benchmarks; 0.6 / 0.3 / 0.1 never beaten after correction.
- **Problems diagnosed:** relation names written differently by different extractors; RQ4 pilot found zero contradictions.
- **Model line-up agreed:** two frontier, one mid, two small answer models, one fixed judge.

earlier result (2 frontier models)	value
P2 vs P0 accuracy	equivalent (+0.000)
P2 tokens vs P0	12% fewer
P3 novelty gate staleness	+12 points
RQ4	blocked

Since then: a pre-run audit fixed eight problems, the final matrix ran (105 cells, 10,360 graded answers), and every follow-up study finished.

Status against the checkpoints: all closed

checkpoint	status	answer
1. best grace period (1 to 21 days)	**done**	no single best: a measured trade-off between history kept and storage (slide 11)
2. best retrieval weights	**done**	0.6 / 0.3 / 0.1 not beaten on a benchmark the search never saw (slide 12)
3. small vs frontier models	**done**	forgetting lifts small models to frontier level; frontier models barely need it (slide 8)
4. revise the thesis document	in progress (me)	Chapters 1 to 3 revised with the professors' comments
5. RQ4 content	**done**	35 multi-agent cells, three topologies, three matching modes (slide 9)
7. other issues found	**done**	8 fixed before the run, 2 after; all in PRE_RUN_AUDIT.md

Fixes that changed what we would have reported: staleness now means asserting the old value, not mentioning it (Claude went from 78% 'stale' to 1%); the judge no longer credits 'I don't know'; reversed facts are now matched; failed extractions are retried.

The final run: five answer models, seven policies, three benchmarks

CONTROLLED Extraction, retrieval (0.6 cosine + 0.3 word overlap + 0.1 recency, top 25), prompt and judge are identical in every cell.

role	model (one pinned provider each)
frontier	GPT-6.1 Sol, Claude Sonnet 5.5
mid	Gemini 3.8 Flash
small	Llama 3.1 8B, Qwen3 14B
extract, judge, P4	DeepSeek V4.1 Flash

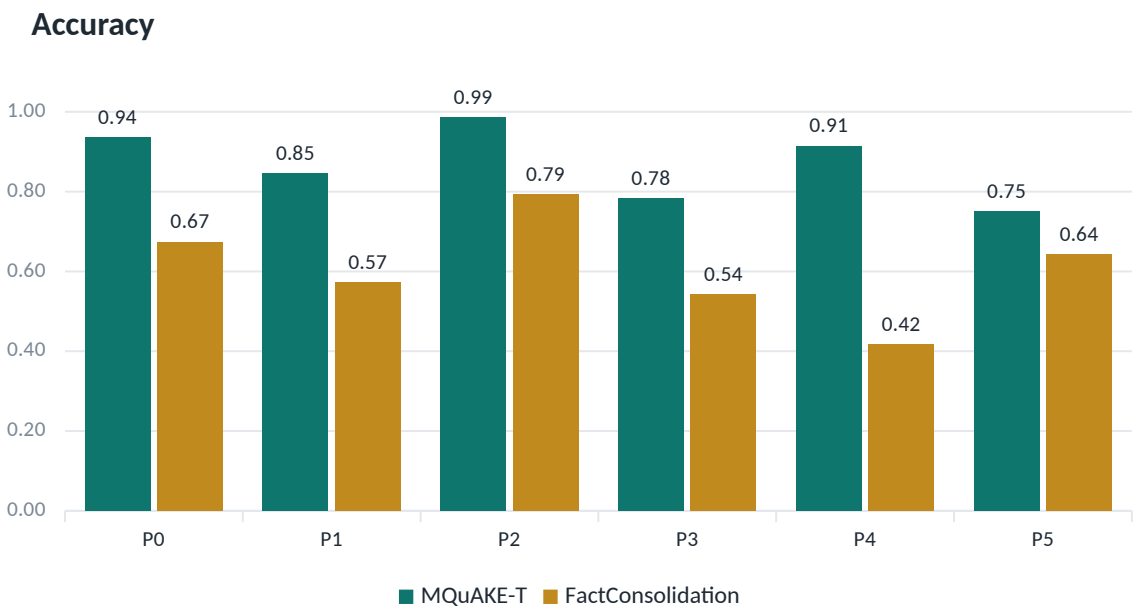
benchmark	questions	kind of change
MQuAKE-T	96	real (Wikidata 2021 to 2023)
FactConsolidation	100	invented on purpose
ConvoMem	100	inside natural dialogue

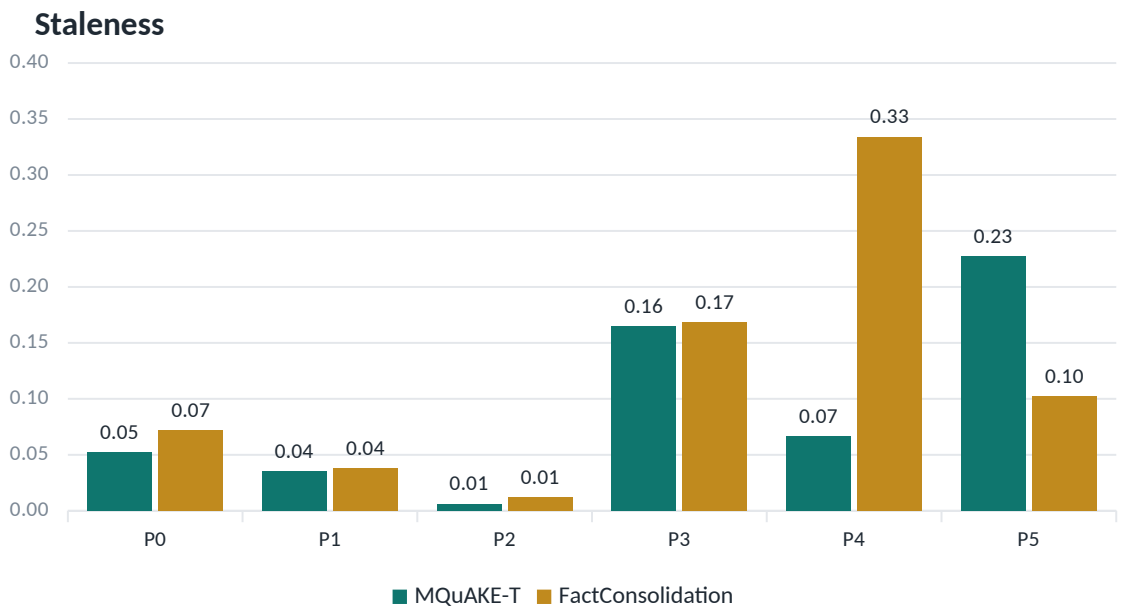
NOMEM	P0	P1	P2	P3	P4	P5
no memory	keep all	delete after 20 days	**hide replaced facts, delete after 7 days**	refuse near-duplicates	LLM reviews memory	keep best-connected value, no dates

Judge check: DeepSeek agreed with Claude Sonnet 5.5 on 96% of 198 answers (Cohen's kappa 0.92: agreement beyond chance, 1 = perfect).

Headline: P2 is the most accurate and least stale policy on both fact benchmarks

STALENESS the share of answers that give the replaced (old) value as the answer. Mean over the five models.





P2 vs P0: accuracy +0.050 on MQuAKE-T and +0.120 on FactConsolidation (both $p < 0.001$, paired McNemar test, Holm-corrected); staleness 0.052 → 0.006 and 0.072 → 0.012.

RQ1: append-only memory puts the old fact in front of the model

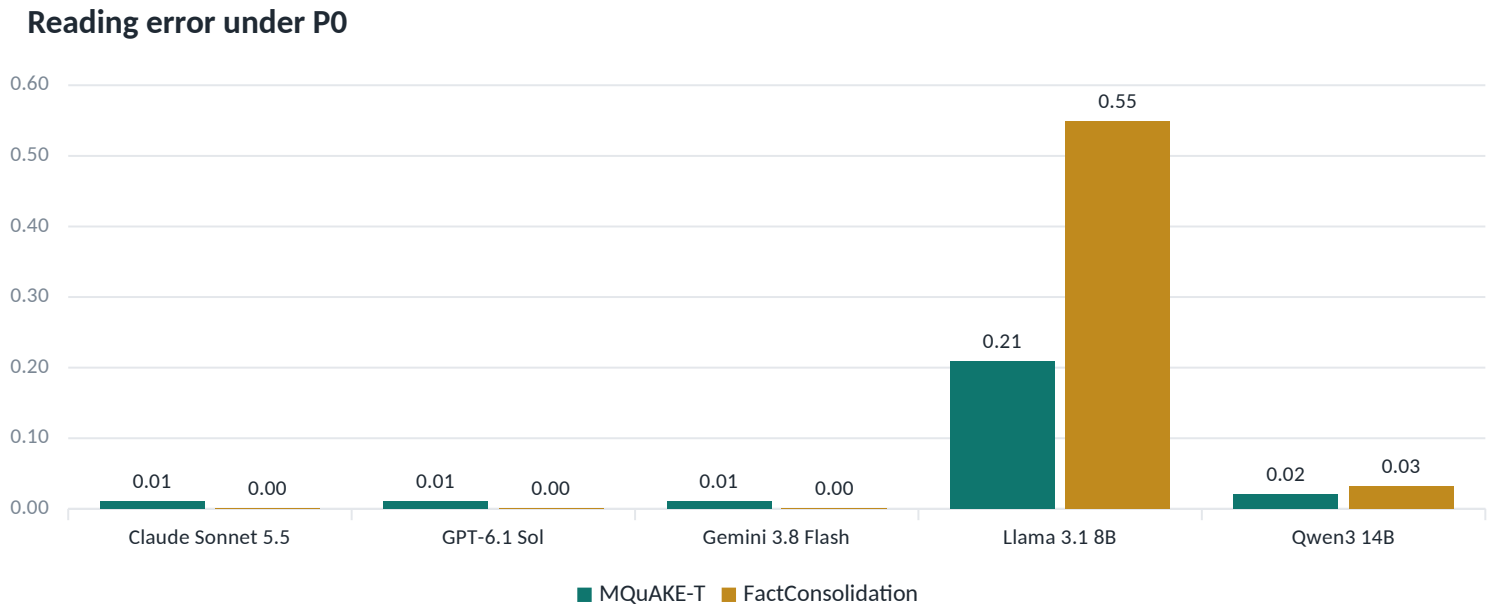
READING ERROR of the answers where the old value was in the prompt, the share that repeated it. It measures the model, not the memory.

100%

of MQuAKE-T questions have the old value in the P0 prompt (95% on FactConsolidation)

7.4 of 25

retrieved facts are already replaced under P0 (MQuAKE-T): about a quarter of the context is outdated



Answer: yes. Retrieval cannot keep the old fact out (old and new differ only in the changed value). Frontier models read the dates correctly; Llama 3.1 8B repeats the old value 21% and 55% of the time.

RQ2: P2 gives the best trade-off on every axis

HOW TO READ each cell is P2 minus the other policy, per question, pooled over models. Every difference shown has p < 0.01 after Holm correction.

P2 minus	MQuAKE-T accuracy	MQuAKE-T staleness	FactConsolidation accuracy	FactConsolidation staleness
P0 keep all	+0.050	-0.046	+0.120	-0.060
P1 time decay	+0.140	-0.029	+0.222	-0.026
P3 novelty gate	+0.202	-0.158	+0.252	-0.156
P4 LLM review	+0.071	-0.060	+0.378	-0.322
P5 topology	+0.235	-0.221	+0.152	-0.090

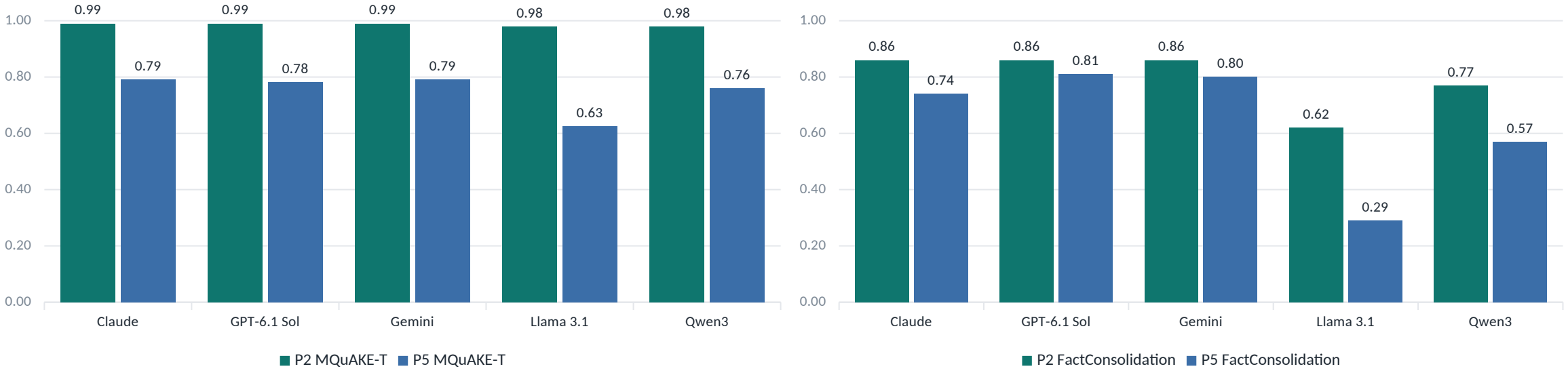
cost, P0 → P2	MQuAKE-T	FactCons.
prompt tokens / q	717 → 577	661 → 601
output tokens / q	98 → 53	164 → 88
facts stored	382 → 337	380 → 320

Why the others lose. P1 deletes true facts that are merely old. P3 rejects corrections as duplicates (31 and 46 writes refused). P4 deletes facts its own model disbelieves: 49 of its 65 deletions on FactConsolidation were the correct, invented answer. P5 keeps the better-connected value, which is usually the old one.

RQ3: it is the timeline that helps, not the graph

P5 sees the same facts and graph as P2 but not the dates, so it must guess which of two conflicting facts is current.

Accuracy, P2 vs P5, by model

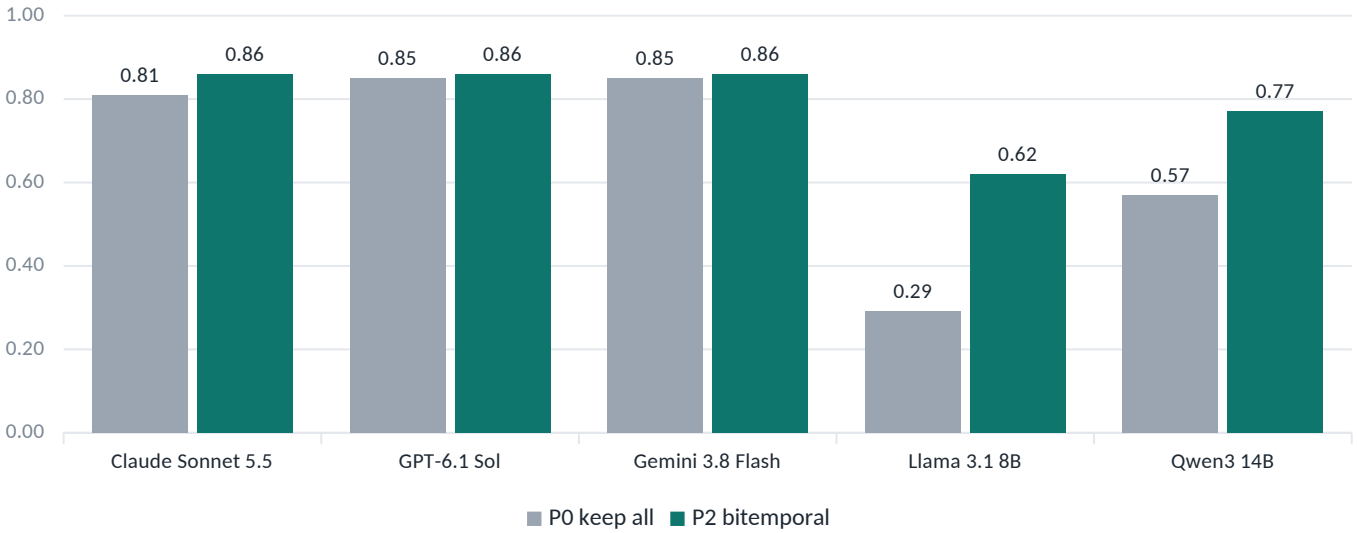


Answer: P2 beats P5 for every model: +0.235 on MQuAKE-T (113 questions won, 0 lost) and +0.152 on FactConsolidation, both $p < 0.001$. Corroboration favours the value mentioned longest, which is the old one.

Checkpoint 3: forgetting matters most for small models

COST PER CORRECT ANSWER list-price USD for 1,000 correct answers under P2 on MQuAKE-T.

Accuracy, P0 → P2 (FactConsolidation)



model	P2 acc. MQuAKE-T	$ / 1k correct
Claude Sonnet 5.5	0.990	$1.73
GPT-6.1 Sol	0.990	$1.10
Gemini 3.8 Flash	0.990	$0.47
Qwen3 14B	0.979	$0.12
Llama 3.1 8B	**0.979**	**$0.01**

Answer: P2 adds +0.12 (MQuAKE-T) and +0.27 (FactConsolidation) accuracy for small models, against +0.01 and +0.03 for frontier models. With P2, Llama 3.1 8B is within one point of the frontier on real updates at about 1/160 of Claude's cost per correct answer.

RQ4: several agents sharing one memory

TOPOLOGY how agents are wired to memory. All answering roles are Gemini 3.8 Flash, so only the memory differs (MQuAKE-T).

topology	who writes	P0 acc. / stale	P2 acc. / stale	facts closed (across writers)
orchestrator	1 extractor	0.990 / 0.010	0.990 / 0.010	n/a
independent	1 per agent, 80% of sessions	0.731 / 0.210	0.733 / 0.210	n/a
shared	5 models, exact names	0.865 / 0.135	0.865 / 0.115	40 (28)
shared	5 models, normalised names	0.865 / 0.135	0.865 / 0.125	60 (46)
shared	5 models, LLM check	0.865 / 0.135	**0.865 / 0.094**	74 (60)

Answer: rankings transfer, Spearman correlation with the single-agent ranking is 0.93 to 0.94 in every topology (1 = same order).

What changes: with five writers, facts are worded differently, and only the LLM check cut staleness (0.135 → 0.094). Independent agents stay at 0.21 stale under every policy: an agent that never saw the update has nothing to forget.

Detecting a contradiction: when a rule is enough, and when to use an LLM

MATCHING how the memory decides a new fact replaces an old one: exact relation names, cleaned-up names, or asking an LLM (as Zep does).

P2 accuracy (Llama / Qwen)	exact	normalised (main run)	LLM check
MQuAKE-T	0.98 / 0.98	0.98 / 0.98	0.99 / 0.98
FactConsolidation	0.60 / 0.76	0.62 / 0.77	**0.69 / 0.85**
ConvoMem	0.06 / 0.08	0.05 / 0.08	0.05 / 0.06

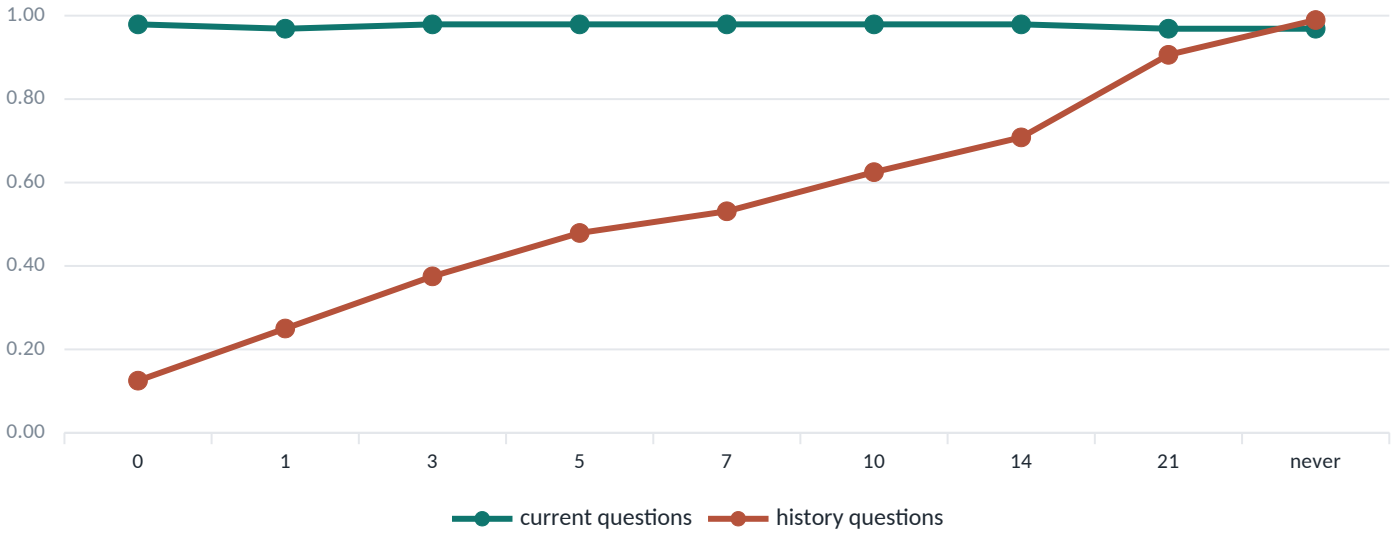
- **Clean names, one extractor:** rules catch 95 of 96 changes.
- **Varied wording:** LLM finds 29 more, +7 to +8 points.
- **Several writers (RQ4):** only the LLM check helps.

Relation to Zep: Zep already closes a contradicted fact instead of deleting it; DRIFT does not claim that idea. DRIFT tests it under control: isolated from the rest of the system, against five other forgetting rules, with staleness measured separately.

Checkpoint 1: the grace period is a trade-off, not a single best value

GRACE PERIOD days P2 keeps a replaced fact before deleting it. History question: 'Before it was last changed, what was the answer to …?'

MQuAKE-T, Qwen3 14B: accuracy by grace period (days)



grace	facts stored	history acc.
0 days	296	0.13
7 days (default)	337	0.53
21 days	373	0.91
never delete	382	0.99

Answer: current accuracy stays 0.97 to 0.98 at every setting; history and storage rise together. Pick grace from how far back users will ask: 0 to 1 day for present-only assistants, the full look-back window (or never delete) for audit, health and finance.

Checkpoint 2: the retrieval weights hold on unseen data

RETRIEVAL SCORE 0.6 x cosine (meaning) + 0.3 x word overlap + 0.1 x recency; the 25 best-scoring facts reach the prompt.

weighting (FactConsolidation, P2)	Llama 3.1 8B	Qwen3 14B
0.6 / 0.3 / 0.1 (thesis setting)	0.62	0.77
0.9 / 0.1 / 0.2 (MQuAKE-T winner)	0.62 (p = 1.0)	0.81 (p = 0.34)
0.8 / 0.1 / 0.1 (ConvoMem winner)	0.56 (p = 0.21)	0.78 (p = 1.0)

- **Search (Week 5):** 1,000 weightings, none better after Holm correction.
- **Held-out (now):** FactConsolidation was never searched; neither winner is significantly better.
- **Holm correction:** adjusts p-values when many weightings are tested, so luck does not look like a win.

What I will claim: not 'the best weights' but a robust setting: not beaten anywhere it has been tested, inside the safe region (recency share at most 0.2), and confirmed on data it was never tuned on.

What did not work, and why: natural conversation

CONVOMEM user-assistant dialogue where a value is stated and later changed, such as a lunch moved from Friday to Saturday.

is the answer in memory?	MQuAKE-T	FactCons.	ConvoMem
current value stored	1.00	1.00	**0.30**
change P2 can act on	0.99	0.92	0.14
P0 accuracy (mean)	0.94	0.67	0.11
P2 minus P0 accuracy	+0.05	+0.12	**-0.04**

- **Extraction ceiling:** the current value is in the dialogue 100 times out of 100 but reached memory 30 times.
- **False replacements:** 14 of the 20 questions P2 lost were wrongly hidden: a second mention of a 'job' or 'course' read as a replacement.
- **LLM matching** removes false closures (281 → 196) but cannot add missing facts.

Takeaway for the thesis: in dialogue, what reaches memory matters more than what leaves it. This is reported as a limitation with its two measured causes, not hidden.

How this strengthened the thesis and the hypothesis

HYPOTHESIS removing only replaced facts gives accuracy at least as good as keeping everything, fewer stale answers, and fewer tokens and less storage.

	before (2 frontier models)	now (5 models, audited)
P2 vs P0 accuracy	equivalent (+0.000)	**better: +0.050 and +0.120, p < 0.001**
who benefits	not measured	**small models: up to +0.33 for Llama 3.1 8B**
staleness metric	counted any mention	counts only asserting the old value
grace period	asserted (7 days)	measured trade-off with history questions
weights	searched on 2 benchmarks	confirmed on a held-out benchmark
RQ4	blocked pilot	**answered: rankings transfer, matching must be semantic**
ConvoMem	inconclusive	negative, with two measured causes

Verdict: hypothesis supported on both fact benchmarks in a stronger form than stated; partly supported on dialogue.

Novelty: what DRIFT adds

POSITION MemGPT, Zep, Mem0 and A-MEM evaluate whole memory systems; DRIFT isolates the forgetting rule and measures it.

- **Forgetting policy as the only variable:** 7 policies x 5 models x 3 benchmarks, everything else fixed.
- **Staleness measured correctly and separately:** asserting vs mentioning the old value reverses model rankings.
- **Mechanism, not only outcome:** retrieval decides if the old value is shown; the model decides if it falls for it.
- **Forgetting as an equaliser:** an 8B model reaches frontier reliability on updates at about 1/160 of the cost.
- **Measured failure modes of popular policies:** novelty gates drop corrections; LLM consolidation reverts updates it disbelieves.
- **Benchmark finding:** LoCoMo and LongMemEval cannot measure memory revision; three that can, audited.
- **Multi-agent memory:** rankings transfer, but contradiction detection across writers needs semantic matching.

What DRIFT does not claim: the invalidate-instead-of-delete mechanism itself (Zep uses it). The contribution is the controlled evidence for when it helps, by how much, and for whom.

Conclusions and next steps

ONE SENTENCE Keep a validity interval on every fact and hide replaced facts at retrieval: more accurate, cheaper, and essential for small models.

- **RQ1:** append-only memory shows the old fact; small models repeat it.
- **RQ2:** P2 is best on accuracy, staleness, tokens and storage.
- **RQ3:** the benefit comes from the timeline, not the graph.
- **RQ4:** rankings hold across topologies; shared memory needs LLM matching.
- **Limits:** dialogue is bound by extraction; rules over-fire on conversational relations.

next	when
Chapters 4 to 6 from FINAL_REPORT.md	this week
Chapters 1 to 3: professors' comments	this week
defence slides from these results	next week

Deliverables: FINAL_REPORT.md (with references), FINAL_RESULTS_DASHBOARD.html (interactive), 12 figures and all tables, PRE_RUN_AUDIT.md.